

ARKO

c.arko.business@gmail.com | Remote, US | [linkedin.com/in/arkchakrabarti](https://www.linkedin.com/in/arkchakrabarti) | github.com/SA-Ark

TECHNICAL SKILLS

Remediation: Production Rescue, Code Remediation, Technical Debt Reduction, Architecture Assessment, Production Hardening, AI-Generated Code Cleanup

Security: OWASP Top 10, SQL Injection Remediation, XSS Prevention, Secrets Detection, Auth/RBAC, API Security

Performance: Lighthouse Optimization, Bundle Optimization, Query Optimization, Caching (Redis), Database Indexing, Code Splitting

Languages: TypeScript, Python, Rust, SQL

Stack: React, Next.js, FastAPI, Node.js, PostgreSQL, Redis, pgvector, Docker, AWS, Jest, Vitest, Playwright, CI/CD

PROFESSIONAL EXPERIENCE

Independent AI Code Remediation Engineer 2024 – Present

Production Rescue — Transforming vibe-coded and AI-generated codebases into production-grade systems

E-Commerce Production Rescue (Online Retailer, \$5M Revenue) — 5 months

- Rescued vibe-coded e-commerce platform: patched 14 critical + 23 moderate OWASP vulnerabilities (SQL injection, XSS, exposed API keys), improved Lighthouse from 32 to 98
- Eliminated N+1 query cascade reducing database queries from 47 to 3 per product page, dropping page load from 8.2s to 1.1s and bundle size from 2.8MB to 340KB
- Built 87% test coverage from 0% (unit, integration, e2e), CI/CD pipeline with automated security scanning, and monitoring stack reducing error rate from 12% to <0.1%
- Phase 2 retained: integrated 6 AI features driving +35% conversion, -60% support tickets, +25% average order value

Enterprise Platform Architecture Overhaul — 4 months

- Took sprawling single-tenant prototype and architected into multi-tenant production SaaS with row-level security, RBAC, Stripe billing, and real-time collaboration
- Remediated: no data isolation between clients, admin routes without auth, zero database indexes, monolithic 15,000-line components

Learning Platform Code Quality Transformation — 3 months

- Replaced prototype AI pipeline (no error handling, no retry logic, no rate limiting) with production-grade system achieving 40% performance improvement
- Built test coverage from 0% to 82% with automated regression testing for AI outputs, serving 10,000+ students reliably with zero unhandled AI failures

Workflow Automation Consolidation — 3 months

- Replaced 5 cobbled-together tools with unified, tested, monitored platform, reducing maintenance burden from 15 hours/week to 2 hours/week

AI Data Specialist — Scale AI 2023 (6 months)

- Evaluated LLMs and performed red-teaming, developing deep understanding of what AI code generators produce wrong and how to systematically remediate AI-generated code

Full-Stack Software Engineer — Contract 2020 – 2023

- Delivered 8 production web applications with focus on performance optimization, security hardening, and production-quality engineering across fintech, healthcare, and e-commerce
- Migrated legacy monolith to microservices for financial services client, reducing deployment time from 3 days to 45 minutes and eliminating 90% of release-related incidents